

Acknowledgments

Thank you who...

This work is supported by my self ahihi. Ok. Now it works fine. Thanks for the help.

中文摘要

摘要

關鍵詞:

Abstract

Named Entity Recognition (NER) in new or specialized domains often requires manually defining entity types and annotating data, a process that is costly and demands domain expertise. This limitation motivates **open-world NER**, where the goal is not only to detect entity mentions from unlabeled text but also to **discover and name latent entity types** automatically. In this thesis, we propose **SAGE**, a semantic-augmented, graph-based framework for open-world entity typing and ontology induction.

SAGE operates without gold entity labels in the target corpus. It first obtains high-recall entity mentions by aggregating predictions from multiple distilled NER teachers, such as GLiNER and UniversalNER, using a small, domain-agnostic label bank solely for span proposal. Each mention is then represented by a **type signature** that combines contextual semantic embeddings, teacher-generated type phrases, and corpus-grounded type evidence mined from definitional and Hearst-style patterns. Using these signatures, SAGE constructs a similarity graph over mentions and applies constraint-aware community detection to induce an initial set of entity type clusters without predefining the number of types.

To address common clustering failures in open-world settings, we introduce a **Critic refinement module** that evaluates cluster quality through learnability. A lightweight classifier is trained to predict cluster assignments, and its confusion patterns and prediction entropy are used to identify over-merged and internally inconsistent clusters, which are iteratively merged or

split in a label-free manner. Finally, SAGE generates human-readable entity type names by mining candidate phrases from clusters and selecting concise, specific labels using an open-source language model.

Experiments on multiple public benchmarks across general, noisy, and biomedical domains demonstrate that SAGE improves type discovery quality and produces more accurate and stable entity class names compared to clustering-only and teacher-only baselines. These results show that SAGE provides a practical and effective solution for automatically constructing entity type inventories when deploying NER systems in new domains.

Keywords: Open-world Named Entity Recognition; Entity Type Discovery; Ontology Induction; Entity Class Name Generation; Graph-based Clustering; Teacher-Augmented Learning; GLiNER; Universal NER; Self-supervised Refinement; Learnability-based Clustering; Domain Adaptation

Contents

List of Tables

List of Figures

Chapter 1

Introduction

1.1 Motivation

Motivation here

1.2 Contributions

Contributions here

- We propose ...
- Experiments show...

Chapter 2

Related Work

2.1 Methodology

2.1.1 Problem Definition

Let D_T be an unlabeled target-domain corpus consisting of sentences or documents. No gold entity spans, entity types, or ontology are provided. The goal is to automatically:

1. detect entity mentions from D_T ,
2. discover latent entity types by grouping mentions,
3. generate a concise, human-readable name for each discovered type.

The system must operate in a fully open-world setting; gold labels are used only for evaluation.

2.1.2 Framework Overview

The proposed framework, **SAGE**, consists of five stages:

1. Multi-teacher mention proposal

2. Type signature construction
3. Graph-based clustering with soft constraints
4. Cluster–Critic refinement
5. Two-stage entity type naming

Each stage is designed to progressively reduce noise while improving semantic coherence and interpretability.

2.1.3 Module A: Multi-teacher Mention Proposal

Goal Obtain a high-recall set of candidate entity spans while avoiding reliance on any single NER ontology.

Label Bank Construction A small, domain-agnostic label bank $\mathcal{L}$ is used only for span proposal, not as target types:

$$\mathcal{L} = \{\text{person, organization, location, event, product, misc}\}$$

These labels are chosen to be coarse, generic, and reusable across domains.

Teachers We employ multiple distilled NER models:

- **GLiNER**: label-bank-driven span detection

- **UniversalNER**: natural-language type prediction
- **GLiNER2** (optional): schema-conditioned extraction

Each teacher proposes spans (m_i) with confidence scores.

Aggregation All spans are merged using a union strategy. Each span is assigned a confidence score based on:

- number of teachers agreeing,
- model confidence,
- heuristics (span length, stopwords, casing).

The output is:

$$M = \{(m_i, x_i, s_i)\}_{i=1}^N$$

where m_i is the span, x_i its context, and s_i its confidence.

—

2.1.4 Module B: Type Signature Construction

Motivation Masked-LM prompts such as “ m is a [MASK]” are brittle and fail across entity categories. We instead construct corpus-grounded, multi-view representations.

View 1: Semantic Span Embedding Each mention is embedded using a contextual encoder:

$$z_i = \text{Encoder}(x_i, m_i)$$

Span representations are computed via token averaging or start–end concatenation.

View 2: Corpus-grounded Type Evidence Type phrases are mined directly from the corpus using linguistic patterns:

- copula: “ X is a Y ”
- apposition: “ X , a Y ,”
- parenthetical: “ X (Y)”
- Hearst patterns: “ Y such as X ”

This yields a phrase set P_i for each mention.

Final Type Signature

$$S_i = [z_i; \text{Embed}(P_i)]$$

This representation supports both clustering and name generation.

—

2.1.5 Module C: Graph-based Clustering with Constraints

Graph Construction We construct a k -nearest-neighbor graph:

- nodes: mentions m_i
- edges: cosine similarity between S_i

Edge weights:

$$w_{ij} = \alpha \cdot \text{sim}(z_i, z_j) + \beta \cdot \text{sim}(P_i, P_j)$$

Soft Constraints Teacher outputs induce pairwise constraints:

- **must-link**: boost edge weight if teachers agree
- **cannot-link**: penalize or remove edge if teachers disagree

Community Detection We apply Leiden or Louvain community detection to obtain initial clusters:

$$\mathcal{C}_0 = \{C_1, \dots, C_K\}$$

without specifying K .

—

2.1.6 Module D: Cluster–Critic Refinement

Core Idea A valid entity type should be *learnable*. Clusters that are difficult to predict indicate semantic inconsistency.

Critic Training A lightweight classifier is trained to predict cluster IDs:

$$f_{\theta}(S_i) \rightarrow \hat{c}_i$$

Diagnostics From the Critic we compute:

- confusion matrix between clusters,
- prediction entropy within clusters,
- constraint violations.

Refinement

- merge clusters with high mutual confusion,
- split clusters with high internal entropy.

The process repeats until convergence.

—

2.1.7 Module E: Entity Type Name Generation

Candidate Mining For each cluster C_k , we collect:

- frequent teacher type phrases,
- frequent mined type phrases,
- salient head nouns.

LLM-based Selection An open-source instruction-tuned LLM selects:

- a 1–4 word noun phrase as the class name,
 - a one-line definition.
-

2.2 Experimental Setup

2.2.1 Datasets

We evaluate on:

- OntoNotes 5.0 (general-domain)
- WNUT-17 (noisy social media)
- BC5CDR (biomedical)

Gold labels are used only for evaluation.

2.2.2 Baselines

- Teacher-only (UniversalNER, GLiNER)
 - One-shot clustering (no refinement)
 - SAGE without graph
 - SAGE without constraints
 - SAGE without Critic
-

2.2.3 Evaluation Metrics

Clustering Quality

- AMI, ARI, NMI
- Purity / Inverse Purity

Name Quality (Core Metric)

- correctness

- specificity
- coherence

Evaluated via human judgment or LLM-as-judge.

—

2.3 Implementation Plan

2.3.1 Minimal Version (Guaranteed Completion)

- GLiNER + UniversalNER span union
- signature = semantic + mined phrases
- graph clustering (single pass)
- LLM naming

2.3.2 Full Version

- add constraints
- add Critic refinement
- naming-aware refinement

—